

Final Exam

Applied Optimization

Procedures:

Leave your **student ID card** on the table during the examination for verification. Write your **first and last name** as well as your **student number** on all answer sheets. Use only black or blue ink pens and write legibly. Exam questions done in pencil or other colored pens will **NOT** be graded. Check if all **18 pages** (9 sheets) are present in your exam copy.

Regulations:

Only one sheet of A4 paper with handwritten notes on both sides is permitted. Course notes, textbooks, other books/printed materials, calculators are **NOT** permitted during the examination. The use of electronic devices such as cell phones, PDAs, laptops, MP3 players, iPods, etc. is strictly **PROHIBITED** during the examination.

The student is deemed to have failed the course if they are found cheating or violating any of the above regulations.

NOTE: You do not necessarily need to finish the whole exam to obtain the maximal grade.

<i>Exercise</i>	<i>Points</i>	<i>Earned points</i>
1. Convex Sets	12	
2. Convex Problems	14	
3. Trust Region	26	
4. Active Set Method	18	
5. Interior Point Method	30	
6. General Questions	10	
Total	110	

1 Convex Sets (12 points)

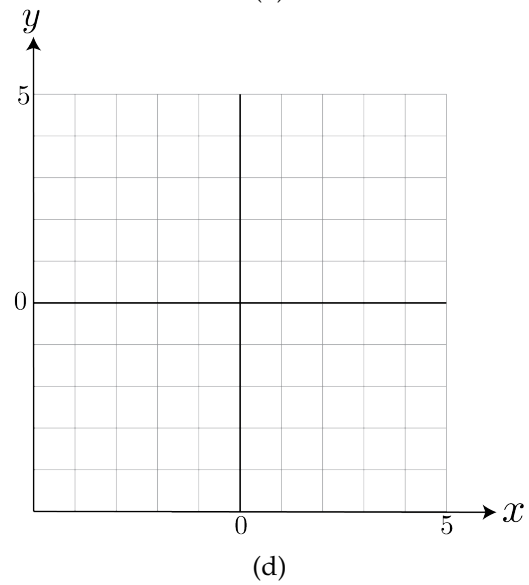
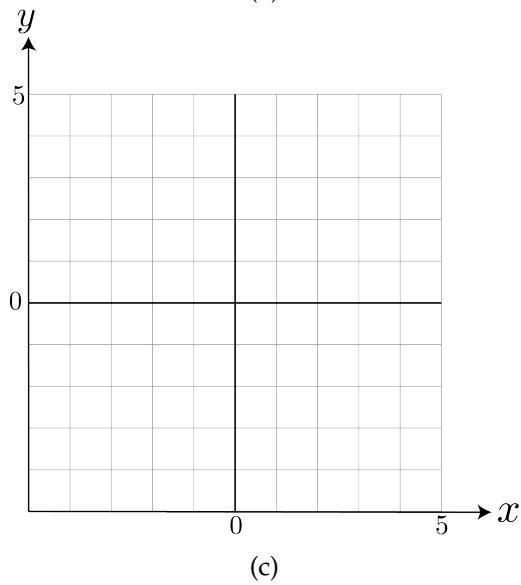
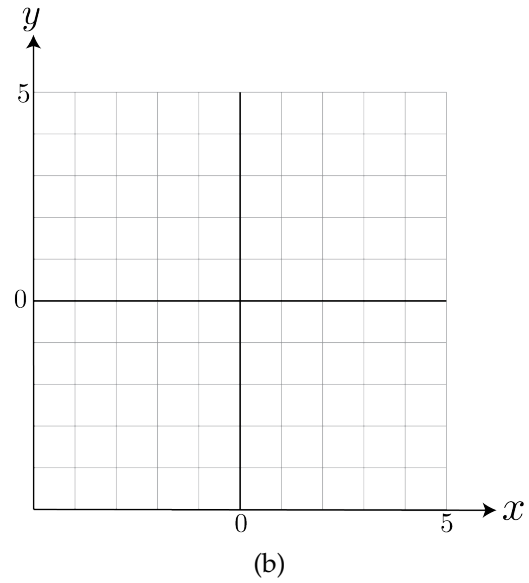
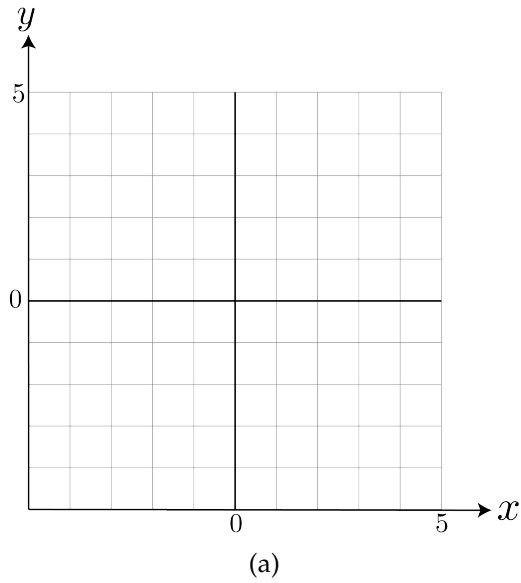
1.1. (10 points): Sketch the following domains and indicate if they are convex or not

$$S_a = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 4, y \leq 1\}$$

$$S_b = \{(x, y) \in \mathbb{R}^2 \mid xy \leq 0\}$$

$$S_c = \{x \in \mathbb{R}^2 \mid \|x\|_1 \geq 1, \|x\|_1 \leq 2\}$$

$$S_d = \{x \in \mathbb{R}^2 \mid x = \alpha(-1, -1)^T + \beta(1, -1)^T + (1 - \alpha - \beta)(1, 1)^T, \\ \text{with } \alpha, \beta \geq 0 \text{ and } \alpha + \beta \leq 1\}.$$



	convex	non-convex
S_a		
S_b		
S_c		
S_d		

- 1.2. (2 points)** Assuming two *convex* sets A and B , is $A \setminus B$ always convex?
If yes, justify your answer with a proof, otherwise, sketch a counter-example.
As a reminder, $A \setminus B = \{x \in A \mid x \notin B\}$

2 Convex Problems (14 points)

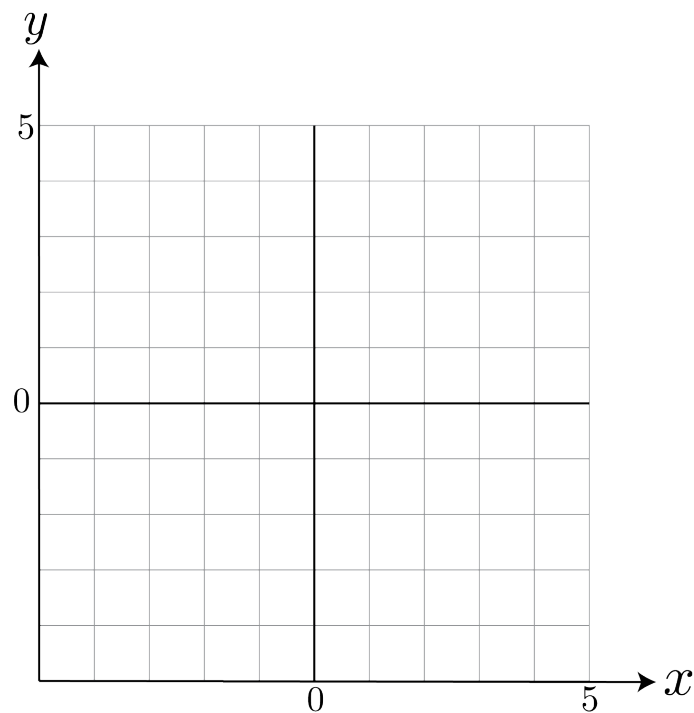
You want to solve the following problem:

$$\begin{aligned} \min \quad & f(x,y) = -(x-1)^2 - y \\ \text{subject to} \quad & x^2 + y^2 \leq 3^2 \\ & x \geq 0 \\ & x + y \geq 0 \end{aligned}$$

where $f : \mathbb{R}^2 \rightarrow \mathbb{R}$

2.1. (2 points) Is the objective function convex? Justify your answer.

2.2. (3 points) Sketch the feasible domain below and mark the solution x^* with a small circle $\circ$.



2.3. (6 points) Verify the optimality of your solution with the KKT conditions.

Note: If you did not answer to the previous question, you can use any feasible point and analyze the KKT conditions at this point.

- 2.4. (3 points) Transform the problem into any constrained, convex problem with the same solution x^* . The feasible domain should be more than a single point and the objective function should be non-constant.

3 Trust Region (26 points)

In this task, we want to minimize the famous Rosenbrock's function:

$$\text{minimize } f(x,y) = (a-x)^2 + b(y-x^2)^2$$

To do so, we will use the Trust Region method with the ∞ -norm ball constraint of radius $\delta_0 = \frac{a}{2}$. This will make each subproblem a quadratic problem with linear constraints. We will thus apply the Cauchy Point method to find the optimal path inside the feasible region of the subproblem.

- 3.1. (2 points):** Rewrite the ∞ -norm ball constraint formulation as a set of 4 linear inequality constraints. As a reminder, the ∞ -norm is defined as $\|(u,v)^T\|_\infty = \max(|u|, |v|)$, with $(u,v) \in \mathbb{R}^2$.

- 3.2. (7 points):** Compute the gradient and the Hessian of the function.

3.3. (3 points): Express the TR subproblem at the starting point $p_0 = (0, 0)^T$.

3.4. (8 points): Compute the Cauchy point of the TR subproblem.

3.5. (6 points): Compute the observed accuracy and argue whether the Trust Region radius δ_0 should be enlarged or shrunk.

In case you failed to compute the Cauchy point, use the following point instead :
 $(u, v)^T = \frac{a}{2}(1, 1)^T$

Note: you can assume in your computations that $a, b > 1$

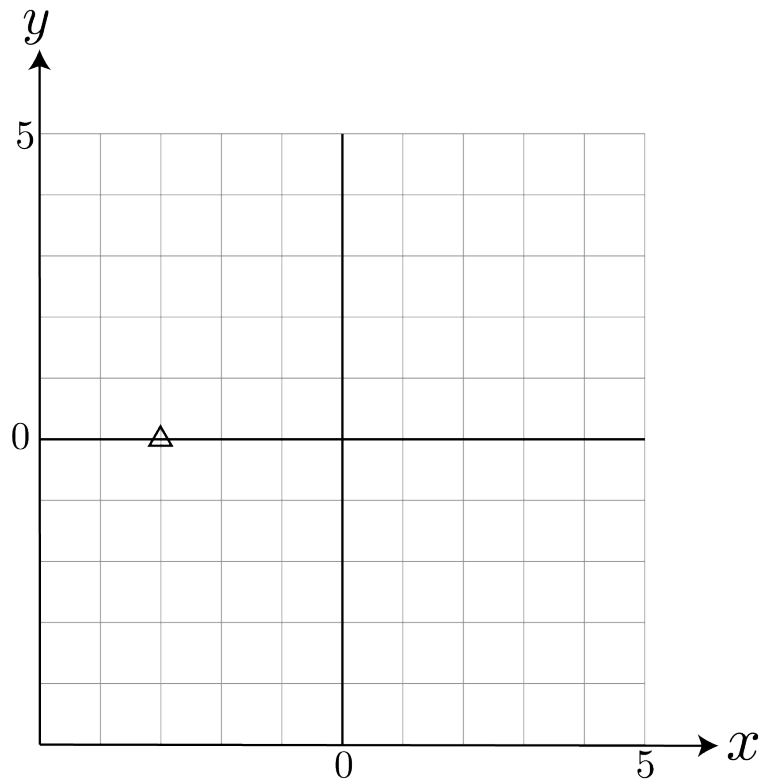
4 Active Set Method with Nonlinear Constraint (18 points)

We want to solve the following problem:

$$\begin{aligned} \text{minimize} \quad & f(x,y) = \frac{1}{2}((x+3)^2 + y^2) \\ \text{subject to:} \quad & x \geq -5 & (C_1) \\ & y \geq -1 & (C_2) \\ & y \leq 5 & (C_3) \\ & y \geq x & (C_4) \\ & (x+5)^2 + y^2 \geq 3^2 & (C_5) \end{aligned}$$

Considering the small number of inequality constraints, you want to use the Active Set Method.

- 4.1. (4 points): Sketch the feasible domain on the grid below. Note that the minimum of the (unconstrained) objective function is already marked with a $\triangle$.



- 4.2. (8 points): Starting from $x_0 = (4,5)^T$, and with the initial set of active constraints $A_0 = \{C_3\}$, draw the optimization path followed by the Active Set Method.
Hint: As you probably noticed, the constraint C_5 is non-linear. Assume that if C_5 is activated, your optimizer can find the minimum constrained to C_5 and follow C_5 exactly.

- 4.3. (6 points): Which part(s) of the algorithm seen in class (below) are not suitable for this particular problem? Identify each problematic part and justify why this is an issue.

Algorithm: Active-Set Method

Input : feasible starting point x_0 , and initial active set A_0

Output: approximate x^*, λ^*

```

for  $k = 0, 1, 2, \dots$  do
    /* Solve KKT System with constraints  $A_k$  to obtain  $x_{k+1}$ ,
        $\lambda_{k+1}$  */
    
$$\begin{bmatrix} Q & B_k^T \\ B_k & 0 \end{bmatrix} \begin{bmatrix} x_{k+1} \\ \lambda_{k+1} \end{bmatrix} = \begin{bmatrix} -q \\ b_k \end{bmatrix}$$

    /* Check whether movement impossible */
    if  $x_{k+1} = x_k$  then
        if  $\lambda_i \geq 0$  for all  $i \in A_k$  then
            | stop with solution  $x^* = x_{k+1}$ 
        else
            /* Remove constraint with most negative  $\lambda$  */
             $j = \arg \min_{i \in A_k} \lambda_{k+1}$ 
             $A_{k+1} = A_k - \{j\}$ 
             $x_{k+1} = x_k$ 
        end
    else
        if there are (inactive) blocking constraints  $\bar{A}$  then
            /* Add first blocking constraint */
             $j = \arg \min_{j \in \bar{A}} (\alpha_j = \frac{b_j - a_j^T x_k}{a_j^T (x_{k+1} - x_k)})$ 
             $x_{k+1} = x_k + \alpha_j (x_{k+1} - x_k)$ 
             $A_{k+1} = A_k \cup \{j\}$ 
        else
            /* Do full step */
             $A_{k+1} = A_k$ 
        end
    end
end
end
```

BONUS (4 points): For any issues that you might have found in exercise 4.3, explain how you would fix it, *for this particular optimization problem*.

5 Constrained Electrons (30 points)

We want to approximate the behaviour of a set of $n \geq 10^6$ electrons, constrained to stay inside of a convex polytope. To keep the problem simple, we will only consider a 2D case, with all electrons lying on the plane. We thus formulate the problem as follows:

$$\begin{aligned} &\text{minimize} && f(p) = \sum_{i,j} E_{ij}, \quad i, j \in \{0, 1, \dots, n-1\}, i \neq j \\ &\text{subject to:} && Ap \leq b \end{aligned}$$

with E_{ij} being the energy of a pair of electrons (in terms of optimization, not Physics), $A \in \mathbb{R}^{m \times 2n}$ and $b \in \mathbb{R}^m$, with m being the number of halfspaces defining the constraining polygon.

The electrons' positions are represented by $p = (p_{0x}, p_{0y}, \dots)^T$, with $(p_{ix}, p_{iy})^T$ being the 2D position of the i -th electron.

5.1. (3 points) As you know, electrons repel each other, no matter the distance. Design a *differentiable* energy E_{ij} that is infinite when two electrons e_i and e_j are at the same position and strictly gets lower as they get further appart.

5.2. (3 points) Formulate this problem as an unconstrained Barrier Method problem.

5.3. (3 points) You want to solve this unconstrained problem using Newton's Method. What condition(s) must be met so this method can be used (assuming Hessian modification is available)? What does this mean for the electrons?
Note: No computations are required for this question.

5.4. (9 points) To solve this problem with Newton's Method, you need to assemble the per-electron Hessian (given by `hessian_f(...)`) into the system Hessian. Fill in the corresponding code below.

Notes: Only set the matrix for the original objective function, NOT for the barrier terms of 5.2.

Hint: Be careful with initialization.

```
using Vec2d = Eigen::Vector2d;
using Mat4d = Eigen::Matrix4d;
using Mat = Eigen::MatrixXd;

//computes the Hessian matrix of the energy between two electrons i and j
//values are stored as
//
// H_ix,ix  H_ix,iy  H_ix,jx  H_ix,jy
// H_iy,ix  ...           :
// H_jx,ix  ...           :
// H_jy,ix  ...           H_jy,jy
//
void hessian_f(const Vec2d& pos_i, const Vec2d& pos_j, Mat4d& H);

//computes the Hessian matrix of the whole system.
//positions are stored as (p_0x, p_0y, p_1x, p_1y,...) and the Hessian values
//should be set accordingly
void hessian_system(const Vec& p, Mat& H){
    /** YOUR CODE HERE **/
}

}
```

- 5.5. (8 points) You are given the code below, that handles both the unconstrained Newton Method algorithm and the Interior Point Method. However, you notice that running this code gives you a solution where all electrons are crammed together around the centroid of the feasible space. Indicate where the error is in the code, how to fix it, and explain why this error would give such a result.

```
// solve with Newton with projected Hessian
Vec solve_newton_PH(Problem *_problem, const Vec& _initial_x,
                    const double _eps, const int _max_iters);

Vec solve_IPM(ElectronsSystem *_obj, const Vec& _initial_x,
              const std::vector<Constraint*>& _constraints, int _max_ipm_iters) {

    double eps(1e-4), mu(10), m(_constraints.size()), t(1.0);
    int _max_newton_iters(1000);

    // log-barrier problem set-up
    // Assume that InteriorPointProblem is a children class of Problem
    InteriorPointProblem ipm_problem(_obj, _constraints);

    Vec x = _initial_x;

    int iter(0);
    while (iter < _max_ipm_iters) {
        ipm_problem.set_t(t);

        x = NewtonMethods::solve_newton_PH(ipm_problem, x, eps, max_newton_iters);

        if (m / t < _eps)
            break;

        t /= _mu;

        ++iter;
    }
    return x;
}
```

- 5.6. (4 points)** Now, suppose that your electrons suddenly become *magic* and that each electron is repelled by at most two other electrons. How would you modify the code of 5.5 and/or 5.4 to drastically improve performance? Assume that, as an input, you have a list of pairs of indices of repelling electrons. You do not need to write any code.

6 General Questions (10 points)

For each statement, mark it as either True or False.

+1 point for each correct answer, -1 point for each wrong answer.

The total is clamped to zero if your score is < 0 .

		T/F
1.	The KKT (Karush-Kuhn-Tucker) conditions provide necessary and sufficient conditions for optimality in nonlinear constrained optimization problems.	
2.	The feasible space defined by a set of linear inequalities can be non-convex.	
3.	The ∞ -norm of an affine function can be optimized with a Linear Program.	
4.	The Lagrange dual problem provides a lower bound on the optimal value of the primal problem.	
5.	In a Linear Program, the objective function and the constraints are allowed to be affine.	
6.	The introduction of slack variables to reduce the inequality constraints is an operation that preserves convexity.	
7.	The Augmented Lagrangian method is a method to optimize nonlinear equality constrained problems.	
8.	The Trust Region method is the best choice for a convex quadratic optimization problem.	
9.	Non-linear optimization problems do not have a unique global minimum.	
10.	All convex functions are differentiable.	